

Gajendra Saraswat

saraswatgajendra97@gmail.com | +919784537496 | +919785845301 | linkedin.com/in/sarques | github.com/sarques | leetcode.com/u/sarques (500+) | saarques.github.io | kaggle.com/sarques

PROFESSIONAL SUMMARY

Software Engineer **3+ years** (Aug 2023 – Present) blending **enterprise backend** (Java/Spring Boot, Python) with **ML/AI** (Scikit-learn, TensorFlow, PyTorch). Delivered 36% faster contract processing, 99% faster portal loads, **4,000+ vulns remediated**, DR for 25+ apps, Kafka resiliency, and **Springer** compression research (5.5x–24.2x). Daily **Windsurf/Devin** with RAG/Agent governance (BNY AI Pioneer L3).

TECHNICAL SKILLS

- **Languages:** Python, Java (8, 17), C++, SQL
- **Data Science & AI:** Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Matplotlib, Keras, OpenCV
- **LLM & AI Engineering:** RAG, Agent Design, Prompt Engineering, Nuggets, Workflow Orchestration, Windsurf, Devin
- **ML Engineering:** Model Dev/Deployment, Feature Engineering, Hyperparameter Tuning, SMOTE, Precision/Recall Evaluation
- **Backend:** Spring Boot, RESTful APIs, Microservices, Apache Kafka, JPA/Hibernate, Flask, ASP.NET (legacy), API Gateway/Manager, Monolithic & Distributed Systems
- **Data & MLOps:** MySQL, PostgreSQL, Redis, Vertica, DB2, Informatica (platform ETL), Docker, Git, Jenkins, GitLab CI/CD, Maven, JFrog Artifactory, Ansible (Regulatory Platform), Google App Engine (Contract Platform), F5 BIG-IP, Cumulus, IIS
- **Quality & Security:** Grafana, Splunk, Orion, PVT, JUnit, Mockito, Selenium, Cucumber, SonarQube, Jira, Confluence, MCP (Model Context Protocol), RBAC, VMAD - Vulnerability Management, SiteMinder

EXPERIENCE

Bank of New York Mellon

Chennai, India

Software Engineer

Aug 2023 – Present

- **Backend & ML Integration (Contract Platform):** Enhanced Spring Boot microservices (@Async/@Transactional) with validation/anomaly detection; 36% faster via parallelization/batching; secure REST/Kafka via **API Gateway/Manager**; deploy via **Google App Engine**, Docker, GitLab CI/CD.
- **Legacy Modernization (Regulatory Platform):** 18.8 MB page 7 min -> 1 min (IIS compression via RDP) -> **5 sec (99%)** via app-wide server-side pagination; full-stack ownership ASP.NET/IIS, DB2/Informatica ETL at platform level.
- **Architecture & Kafka Resiliency:** DR for **25+ apps** multi-cluster enabling automated failover; led Java 17/Angular 17 zero-downtime upgrades and migrated 30+ business-critical reports preserving entitlements. Validated **Kafka backup failover (8 topics, 2 clusters, earliest/latest offsets, lag)** Cluster1->Cluster2 pre-prod. Migrated **Windows 2016->2022 (QA 3->2, PROD 3->3)** for Regulatory Platform (DB2/Informatica ETL at platform level) via Ansible, **F5/Cumulus** isolated Chrome validation, Grafana/Splunk/Orion/PVT, zero downtime.
- **Security & Responsible AI:** Remediated **4,000+ (Regulatory: 28 – 5 Critical, 8 High)** securing auth (SonarQube, VMAD); **RBAC + go-to for entitlements** (Access Hub prod, direct lower-env provisioning); audited **Devin CoT secret exposure** (chat/GitLab), rotated secrets, authored guidelines (AI Pioneer L3).
- **Data-Driven Quality & AI Tooling:** Python metrics/dashboards + **NextGen 300+ / 95%** -> fully automated regression (auto-gen -> Jira via MCP). Fixed **P1 MIME outage (web.config text/html wrapping)** with SiteMinder + regulatory report pre-deadline. Daily **Windsurf/Devin** (RAG/Agent) for code-gen/debugging across ASP.NET/IIS.

SELECTED PROJECTS

- **Credit Card Fraud Detection** Sep 2020 – github.com/sarques/credit-card-fraud-detection (16 stars/16 forks): Logistic Regression (balanced weights) **97.6% acc, 89.6% recall**, SMOTE for imbalance, Flask/REST, Docker/Heroku.
- **Leukemia Detection (PyTorch/SeResNext)** Jun - Sep 2019 – Blood-Cancer-Diagnosis-API: **90%** accuracy on microscopic images, Pandas/NumPy preprocessing, Docker Hub hub.docker.com/r/sarques/bcpmodel (173 pulls).
- **Autoencoder-Huffman Compression** Dec 2024 – Springer 5.5x–24.2x: Encoder-decoder + Huffman, MNIST/CIFAR-10/Fashion MNIST. AI-Without-Borders

EDUCATION

- **NIT Warangal** – M.Tech Computer Science, CGPA 7.13 Sep 2021 – Jun 2023
- **MBM Jodhpur** – B.E. Computer Science, CGPA 7.48 Aug 2016 – Oct 2020

PUBLICATION

- **Autoencoder-Huffman Ensemble Model for Image Compression** Dec 2024
Springer, Singapore – link.springer.com/chapter/10.1007/978-981-97-6714-4_18
Proposed two-stage autoencoder-Huffman approach achieving **5.5x–24.2x** compression ratios on MNIST, CIFAR-10, Fashion MNIST with minimal information loss.

HONORS & AWARDS

- **GATE 2021:** Secured **99.1 percentile (AIR 943)** in Graduate Aptitude Test in Engineering (Computer Science)
- **Kaggle SIIM-ISIC Melanoma Classification:** Ranked in **top 19% globally** in large-scale medical imaging competition

CERTIFICATIONS

- **AI Pioneer – Proficiency Level 3:** BNY Internal AI Learning Program – RAG, Agent Design, Prompt Engineering, Nuggets, Workflow Orchestration, AI Governance
- **Machine Learning** – Coursera
- **Deep Learning Specialization** – Coursera
- **Introduction to Data Science in Python** – Coursera
- **Statistical Thinking in Python (Part 1)** – DataCamp
- **Mathematics for Machine Learning** – Coursera